

Ethics, Privacy, Security, Bias, and Responsible Data Use

DAEN 459 - Fall 2026
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Source map: governance and ethics

- **NIST AI RMF:** AI risk, trustworthiness, lifecycle governance
<https://www.nist.gov/itl/ai-risk-management-framework>
- **NIST Privacy Framework:** privacy risk management
<https://www.nist.gov/privacy-framework>
- **ACM Code of Ethics:** professional duties for computing work
<https://www.acm.org/code-of-ethics>
- **OECD AI Principles:** trustworthy AI and human-centered values
<https://oecd.ai/en/ai-principles>

Source map: security, law, and data stewardship

- **OWASP Top 10:** common web application security risks
<https://owasp.org/www-project-top-ten/>
- **GDPR Article 5:** lawfulness, fairness, transparency, minimization
<https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- **FAIR Principles:** findable, accessible, interoperable, reusable data
<https://www.go-fair.org/fair-principles/>

Source map: documentation practices

- **Datasheets for Datasets:** document dataset motivation, composition, collection, preprocessing, uses, and risks
<https://arxiv.org/abs/1803.09010>
- **Model Cards:** document model purpose, evaluation, intended use, limitations, and ethical considerations
<https://arxiv.org/abs/1810.03993>

How to use these sources in capstone projects

- Use **NIST AI RMF** to identify and govern AI-related risks.
- Use **NIST Privacy Framework** and **GDPR principles** to justify privacy controls.
- Use **OWASP Top 10** to evaluate application and API security.
- Use **FAIR principles** to improve data stewardship.
- Use **Datasheets** and **Model Cards** as required documentation templates.

Why responsible data engineering matters

- Data pipelines create real-world consequences.
- Ethical failures often begin as design shortcuts.
- Responsible systems require technical, legal, and social controls.

Core responsibilities

- Collect only what is needed.
- Protect data across its lifecycle.
- Document provenance, consent, assumptions, and limitations.
- Test for bias, leakage, misuse, and downstream harm.
- Make systems auditable and governable.

Privacy

- Data minimization
- Purpose limitation
- Consent and notice
- De-identification and re-identification risk
- Retention and deletion policies

Security

- Authentication and authorization
- Encryption in transit and at rest
- Secrets management
- Least privilege
- Logging, monitoring, and incident response

Bias and fairness

- Historical bias
- Sampling bias
- Label bias
- Measurement bias
- Deployment feedback loops

Responsible data lifecycle

1. Define purpose and stakeholders.
2. Assess data sources and permissions.
3. Build secure pipelines.
4. Validate quality and representativeness.
5. Document assumptions and risks.
6. Monitor performance and harms after deployment.

Project risk checklist

- Who could be harmed by errors?
- What sensitive attributes are present or inferable?
- Could the data be re-identified?
- Who has access to raw data?
- Are outputs explainable and contestable?
- What happens when the model or pipeline fails?

Documentation artifacts

- Data dictionary
- Data lineage diagram
- Dataset datasheet
- Security threat model
- Bias and fairness assessment
- Retention and deletion policy
- Model card, if ML is used

Capstone deliverables

- Architecture diagram
- Data governance plan
- Privacy and security controls
- Bias-risk analysis
- Responsible-use statement
- Monitoring and rollback plan

Discussion

- What should this system refuse to do?
- What data should never be collected?
- What assumptions could fail in production?
- Who audits the system?
- How would users challenge an incorrect result?

Team Homework: Responsible Data Use Review

Goal for next Friday lab: prepare a 3–5 minute team presentation on your project's ethics, privacy, security, bias, and responsible data-use risks.

Analyze your project

- What data do you collect, store, or process?
- What privacy or security risks exist?
- What bias or fairness concerns could arise?
- Who could be harmed by incorrect results?

Present your plan

- Project context and data used
- 2–4 top responsible-use risks
- Concrete design or governance responses
- One open question for class feedback

Use at least two references: NIST AI RMF, NIST Privacy Framework, ACM Code of Ethics, OWASP Top 10, OECD AI Principles, GDPR, FAIR, Datasheets for Datasets, or Model Cards.

Guiding question: What could go wrong with our project, and what are we doing about it?